

Notes on Linear Factor Models [Lab3]

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1 Introduction

In quantitative finance, separating α (excess returns generated by skill) from β s (returns driven by systematic risk factors, e.g. market risk) is of paramount importance. Most of an asset's return is driven by ancillary factors (such as the market, sector or style, e.g. momentum, value, size, etc.) that an investor may not have a specific view on. To isolate the idiosyncratic return, we must model and strip away the systematic risk.

Whether characteristic or statistical, factor models share a common mathematical backbone. We model the excess returns of n assets at time t as:

$$r_t = \alpha + Bf_t + \epsilon_t$$

Where:

- r_t is the $n \times 1$ vector of asset excess returns (in excess of a risk-free rate);
- α is the vector of expected idiosyncratic returns;
- B is the $n \times m$ matrix of *factor loadings* (betas), representing asset sensitivity to factors;
- f_t is the $m \times 1$ vector of *factor returns* (systematic drivers);
- ϵ_t is the $n \times 1$ vector of *idiosyncratic* (residual) returns.

The covariance matrix of returns, Σ , is decomposed into a low-rank systematic component and a diagonal (strict factor model) or sparse (approximate factor model) idiosyncratic component:

$$\Sigma = B\Sigma_f B^T + \Sigma_\epsilon$$

Question 1 *How many parameters do we estimate when using this model? How does this number compare to the sample covariance matrix?*

2 Characteristic Factor Models

Characteristic models, also called fundamental factor models, are the industry standard. They assume that observable economic phenomena explain the cross-section of returns.

2.1 Methodology

In this framework, the loadings matrix B is known, and we estimate the factor returns f_t .

Inputs: We construct B using asset characteristics. These include:

- *Binary Factors:* Dummy variables for country or sector (e.g. energy, technology, etc.) membership;
- *Style Factors:* Continuous variables such as momentum (price trends), value (fundamentals like book-to-price), or size (market capitalization).

Question 2 *How do we estimate the missing pieces of this model?*

2.2 Practical Implementation Challenges

1. **Data Cleaning (Winsorization):** Raw financial data contains outliers that can skew least-squares estimation. It is standard practice to winsorize (clip) returns and characteristics to symmetric percentiles (e.g., 5th and 95th) to reduce variance and improve robustness.
2. **Collinearity Constraints:** If we include a market factor (all 1s) and exhaustive sector dummy variables (which sum to 1), the matrix B becomes rank-deficient. We must apply constraints, such as forcing sector betas to sum to zero, to obtain a unique solution.
3. **Standardization:** Style factors must be normalized (z-scored) cross-sectionally to ensure they are comparable and to remove scale dependencies.

3 Statistical Factor Models

Statistical models assume we know neither the loadings B nor the factor returns f_t . We infer both simultaneously purely from the covariance (or correlation) structure of returns. We typically use the covariance matrix when the assets are homogeneous and have similar variances, while we use the correlation matrix with heterogeneous assets, e.g. in a multi-asset portfolio.

3.1 Methodology

The simplest and most used statistical factor model relies on Principal Component Analysis (PCA). In this setting, we start with an estimate of the covariance matrix $\hat{\Sigma}$ and state the problem as:

$$\max_{\mathbf{w}_1} \mathbf{w}_1^T \hat{\Sigma} \mathbf{w}_1, \quad \text{s.t. } \|\mathbf{w}_1\| = 1$$

$\mathbf{w}_1$ is called the first principal component and financially it is the maximum variance portfolio where the sum of the squared notionals is equal to 1. Then we iteratively impose that the new direction (portfolio) we are looking for is orthogonal to all the previously computed principal components, e.g. for the second principal component we would solve:

$$\max_{\mathbf{w}_2} \mathbf{w}_2^T \hat{\Sigma} \mathbf{w}_2, \quad \text{s.t. } \|\mathbf{w}_2\| = 1, \quad \mathbf{w}_2^T \mathbf{w}_1 = 0$$

It is well known that the solution to this problem is given by the eigenvectors of $\hat{\Sigma}$, i.e. $\mathbf{w}_i = \mathbf{u}_i$, where $\hat{\Sigma} \mathbf{u}_i = \lambda_i \mathbf{u}_i$.

Question 3 *What does λ_i represents?*

The largest eigenvalues of the return covariance matrix correspond to the pervasive factors (systematic risk), while the remaining bulk of small eigenvalues corresponds to noise (idiosyncratic risk).

To recast the problem in the usual linear factor framework, we think of PCA as a best low-rank approximation. In matrix form we can write:

$$R = BF + E$$

- $R \in \mathbb{R}^{n \times T}$ is the matrix of de-meaned asset returns;
- $B \in \mathbb{R}^{n \times m}$ is the matrix of factor loadings (exposures);
- $F \in \mathbb{R}^{m \times T}$ is the matrix of factor returns;
- E represents the idiosyncratic (unexplained) returns.

We seek to find the loadings B and factor returns F that minimize the total unexplained variation (the residuals) solving the following optimization problem:

$$\min_{B, F} \|R - BF\|_F^2 \quad \text{s.t. } \text{rank}(BF) \leq m$$

where $\|\cdot\|_F$ is the Frobenius norm. This is equivalent to finding the best low-rank approximation $\hat{R}$ of the return matrix R :

$$\min_{\text{rank}(\hat{R}) \leq m} \|R - \hat{R}\|_F^2$$

Using the Singular Value Decomposition (SVD) $R = USV^T$, with U, V orthonormal and S diagonal, the solution is given by $\hat{R} = U_m S_m V_m^T$, where U_m and V_m contain the first m columns of U and V , and S_m contains the largest m singular values.

The relationship to the factor model is established by setting:

$$B = U_m \tag{1}$$

$$F = S_m V_m^T \tag{2}$$

It is then sufficient to note that $\hat{\Sigma} = \frac{1}{T-1} R R^T = \frac{1}{T-1} U S V^T V S U^T = \frac{1}{T-1} U S^2 U^T = U \Lambda U^T$, where $\Lambda = \frac{S^2}{T-1}$.

Question 4 *Can you suggest another, equivalent, solution?*

Question 5 *Performing PCA on the covariance matrix is equivalent to performing SVD on the demeaned asset returns matrix. What can you tell about PCA on the correlation matrix?*

3.2 Key Challenges in Statistical Models

While statistical models are powerful because they require no external fundamental data, they introduce specific complexities:

1. **Interpretability:** Unlike characteristic models, PCA factors do not have inherent names (e.g., "value"). They are simply orthogonal vectors maximizing variance. However, the first factor is almost always the market, but interpretability remains a key issue.
2. **Eigenvalue Bias:** Empirical eigenvalues are biased. Large sample eigenvalues are biased upwards (overestimating risk), and small ones are biased downwards. Shrinkage techniques are required to correct this.
3. **Instability (Turnover):** Eigenvectors are identified only up to a sign (direction). From one day to the next, a factor loading vector might flip sign or rotate slightly, causing massive apparent turnover in the factor portfolio. We must algorithmically align eigenvectors between periods to ensure stability.

References

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